Tomato Fungal Diseases

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Damping off - Pythium spp, *Phytophthora, Fusarium* and *Rhizoctonia solani*

- An important disease especially in tomato and also other vegetable crops like bell pepper, chillies, brinjal, cabbage, cauliflower, broccoli etc. through the world in **temperate and tropical climates**
- Disease severity was high **in forest and heavy soils**
- The disease is fairly common in poorly managed nurseries and green houses, incited often by the seed and soil borne pathogens.

Damping off - Pythium spp, *Phytophthora*, *Fusarium* and *Rhizoctonia solani*

- Affects seeds, seedlings and roots of plants
- It results poor seed germination and more seedling mortality in nursery but also carried to the main field.
- Damping-off typically occurs when old seeds are sown in wet soil in cool weather and incidence increased with ill drained conditions

Damping off - Pythium spp, Phytophthora, Fusarium and Rhizoctonia solani

- Losses vary with soil moisture, temperature and other factors
- The loss is mostly due to **pre emergence damping off**
- Losses due to damping off is estimated as 25-75%
- Seedling mortality may be as high as **90%**

Damping off - Pythium spp, *Phytophthora*, *Fusarium* and *Rhizoctonia solani*

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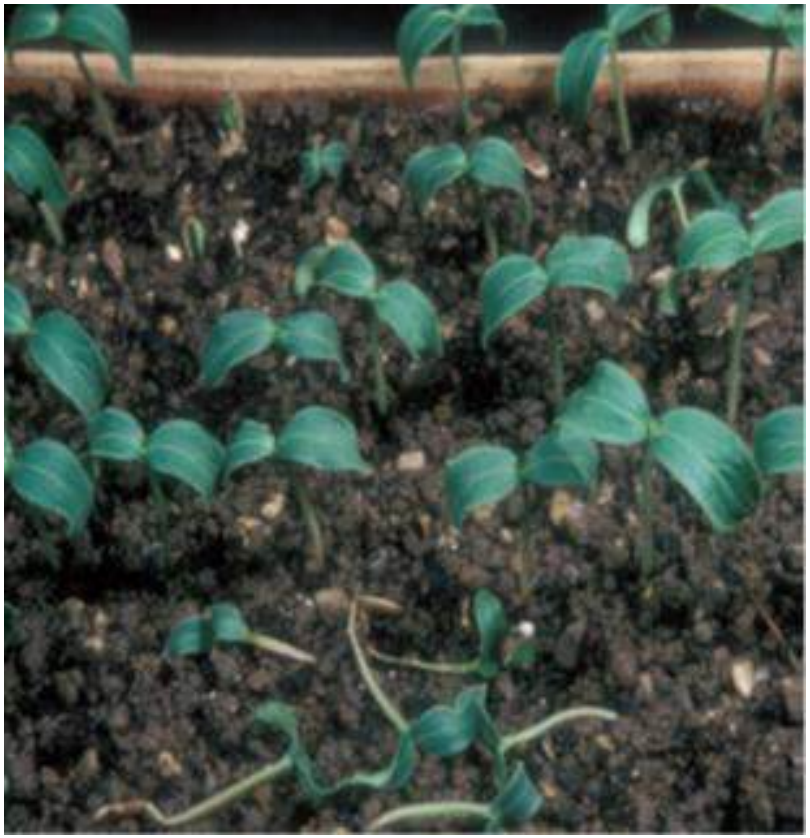

Damping off - Pythium spp, *Phytophthora, Fusarium* and *Rhizoctonia solani*

Greater damage is to seeds, seedling roots during germination either before or after emergence

Symptoms- two phases

i) Pre-emergence damping-off

ii) Post-emergence damping-off

1. Pre-emergence damping off:

- Seeds may rot even before the seed coat breaks become soft, mushy and turn brown, shrivel and finally disintegrate
- Complete rotting of radicle and plumule of the seed below the soil surface, **there by the poor stand**
- Seedlings are killed before they reach the soil surface

Damping off - Pythium spp, *Phytophthora*, *Fusarium* and *Rhizoctonia solani*

Post-emergence damping off :

In young seedlings after emergence

- Due to infection the roots and stems at or below the soil line become water soaked, turns brown, soft

- Infected stems becomes softer and shrivel become much thinner (Wire stem symptoms) than the above and results in collapse of seedlings, which withers quickly and dies.

Damping off - Pythium spp, *Phytophthora, Fusarium* and *Rhizoctonia solani*

Symptoms contd:

In older plants

- Lesions are produced on infected roots and stems , the plants become stunted, wither and die
- Disease appear in patches both in nursery and field
- Disease is more severe when the soil moisture is medium to high (50%water holding capacity or more) with high temperature which is not favourable for the growth of the seedlings

Damping off - Pythium spp, *Phytophthora*, *Fusarium* and *Rhizoctonia solani*

- Environmental conditions such as high humidity levels, cool soil, rich potting soils, and planting too deeply will also encourage its growth.

Spread

- Through irrigation water, implement during cultural operations etc.

Damping off - Pythium spp, *Phytophthora*, *Fusarium* and *Rhizoctonia solani*

Pathogen (s):

- Species of *Pythium*, *Phytophthora*, *Fusarium* and *Rhizoctonia solani* are the most commonly associated pathogens.

- Among the fungi, the species of *Pythium* viz., *P. butleri* and *P. aphanidermatum* are more common than others.

- The mycelium of *Pythium* is intracellular, consists of slender, coenocytic and much branched hyphae.

Damping off - Pythium spp, *Phytophthora, Fusarium* and *Rhizoctonia solani*

Pathogen (s):

Asexual reproduction:

- Mostly by means of zoospores, that which are differentiated in a vesicle, kidney shaped and biflagellate, which later become encysted and germinate by forming germ tube.

Sexual reproduction:

- One male nucleus from antheridium passes into the oogonium through conjugation tube which then forms a thick wall spores known as oospore.

Damping off - Pythium spp, *Phytophthora*, *Fusarium* and *Rhizoctonia solani*

Disease cycle and epidemiology

- Most of the fungi responsible for this disease are both seed and soil borne
- The species of *Pythium* survives from one season to the other as mycelium in plant debris or oospores in the soil.
- *Rhizoctonia solani* survives in soil indefinitely in the form of sclerotia.
- Hyphae grow from these sclerotia and infect seedlings in the nursery beds.
- Species of *Pythium* attack the plants when they are in juvenile stage.

Damping off - Pythium spp, *Phytophthora*, *Fusarium* and *Rhizoctonia solani*

Disease cycle and epidemiology

- Soil moisture near saturation stimulates mycelial growth as well as asexual reproduction.
- High soil moisture, pH 6.0 and heavy soils favour disease development.
- **Pre-emergence damping-off** is maximum at **20-25°C while post emergence at 30- 40°C.**
- The disease is further **aggravated in ill-aerated soils with poor drainage when seedlings stand is more**.

Damping off - Pythium spp, *Phytophthora, Fusarium* and *Rhizoctonia solani*

Management

- Change the nursery site every year.
- Soil solarisation of the bed with transparent polyethylene or polyvinyl (25 µm) sheet for 40-45 days during summer months **or** treat the soil with Formalin (5%) at least 20 days before sowing **or** apply bioagents like *Trichoderma harzianum* **or** *T. viride* (40 g/m2).
- Avoid over crowding of seedlings by using recommended seed rate
- Treat the seed with captan (0.3%).

Damping off - Pythium spp, *Phytophthora, Fusarium* and *Rhizoctonia solani*

Management contd…

- After seedling emergence from the soil, drench the bed with the mixture of mancozeb (0.25%) and carbendazim (0.1%) and repeat at 7-10 days interval.
- Give light but frequent irrigations.
- Drenching with 1% Bordeaux mixture or COC@ 0.3% or metalaxyl @ 0.2%

Tomato late blight (*Phytophthora infestans)*

- This disease is present in all the tomato growing areas of the world
- In India first reported from Himachal Pradesh, subsequently from Karnataka, Haryana and Punjab
- Later in Andhra Pradesh, Telangana and Tamilnadu
- It was considered as a serious threat until 2008 in tomato , however it occurred in epidemic form from July to December, 2009 and 2010 in Karnataka

Tomato late blight (*Phytophthora infestans*)

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Tomato late blight (*Phytophthora infestans*)

Symptoms

- On leaves, it appears as pale, green, irregular spots on the tips and margins, which in moist weather enlarge rapidly with central tissue turning necrotic and dark brown or black.
- On the lower surface of the leaves, a white downy growth of the fungus appears around the dead areas, which is more prevalent in morning hours when climate is moist.
- **Brown streaks along the stems** also.
- On tomato fruits, dark olivaceous greasy appearing spots are formed, which gradually cover the entire fruit surface
- Finally rotting of leaves and fruit takes place.

Tomato late blight (*Phytophthora infestans)*

Management

- Follow crop rotation and avoid solanaceous crops like potato in rotation.
- Improve drainage of the field and keep field free from weeds.
- Collect and destroy the infected fruits regularly.
- Before the initiation of the disease, apply one protective spray of mancozeb (0.25%) followed by **metalaxyl + mancozeb (0.25%)** at critical stage of disease appearance and repeat the sprays at 7-10 days interval.
- Cultivation of resistant F1 Hybrid: Arka Abhed

Tomato buck eye rot (*Phytophthora nicotianae* var. *parasitica*)

- This disease is present in all the tomato growing areas of the world
- In India first reported from Himachal Pradesh, subsequently from Karnataka, Haryana and Punjab
- The **fruit rot** becomes more devastating under **high humidity and wet soils.**
- Losses due to this disease ranges from **35-40%** and fruit rot incidence **upto 90%** under high humidity and good rainfall conditions
- The pathogen has been reported to attack the plants at all growth stages causing damping off, collar rot and stem canker, blossom blight and buck eye fruit rot

Tomato buck eye rot (*Phytophthora nicotianae* var. *parasitica*)

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Tomato buck eye rot (*Phytophthora nicotianae* var. *parasitica*)

Symptoms

- **Pale to dark elongated spots** may appear **on the collar portion** of the plants extending 15-20 cms above the ground there by **girdling the stem** completely **which is uncommon.**
- The diseased stem is **spongy to touch**
- The tops of such **branches wilt suddenly and die**
- **Blossom blight** may occur during the prolonged period of heavy rainfall
- The fungus attacks **petals, sepals, ovary and stamens causing rotting and shredding of flowers**

Tomato buck eye rot (*Phytophthora nicotianae* var. *parasitica*)

Symptoms

- **Fruit rot is more serious during monsoon season**
- **Immature fruits (green colour) either in contact or near the soil** level irrespective of their development stages are susceptible .
- Water soaked light brown discoloured spots appear which increase readily showing **concentric dark brown rings slightly resembling the markings as a buckeye nut**
- Subsequently the disease spreads to upper fruits
- The **lesions rapidly enlarge and with in 3-4 days,** whole of the fruit surface turns **dark brown and soft to touch.**
- Later, these fruits become **mummified and drop off** from the plant.

Tomato buck eye rot (*Phytophthora nicotianae* var. *parasitica*)

Disease cycle

- The fungus overwinters in the soil in the form of **oospores or chlamydospores and can remain active in soil for at least one year** without the support of a susceptible host
- With the onset of monsoon rains, in the presence of high soil moisture and moderate temperatures (20-25°C), the **chlamydospores and oospores** start germinating by producing mycelium and sporangia.
- In warm and humid weather, white **flocculent superficial growth of the fungus** consisting of sporangia and sporangiophores also develops **on the diseased fruits.**

Tomato buck eye rot (*Phytophthora nicotianae* var. *parasitica*)

Disease cycle cond…

- The **sporangia in turn produce biflagellate zoospores**, which are splashed by rain and cause secondary infection..

Tomato buck eye rot (*Phytophthora nicotianae* var. *parasitica*)

Epidemiology

- The symptoms develop on fruits after 3-4 day of infection.
- Maximum fruit infection under field conditions occurs at a temperature range of 20- 25°C, RH > 80 per cent and high rainfall conditions.
- Higher doses of N resulted in higher fruit rot while higher levels of P resulted in more yield of healthy fruits and less fruit rot.

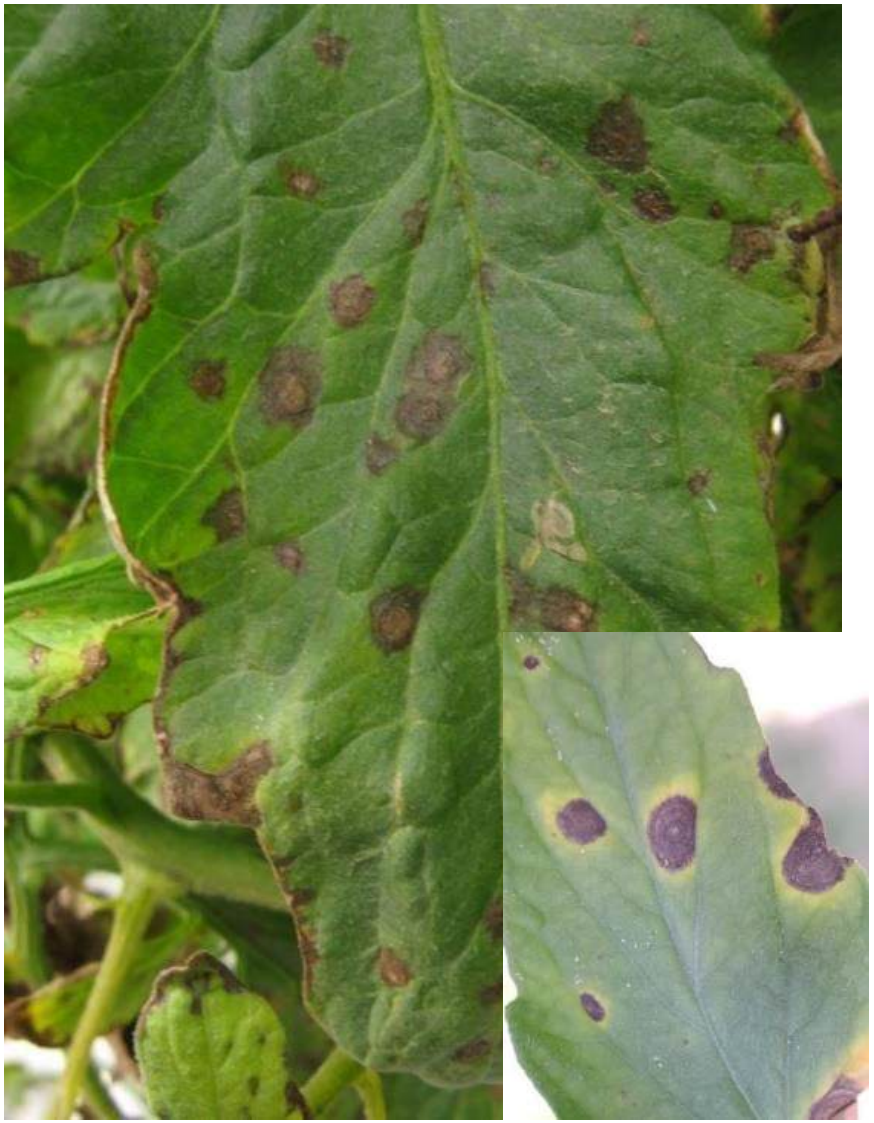
Tomato buck eye rot (*Phytophthora nicotianae* var. *parasitica*)

Management

- Wider spacing should be adopted
- Stake the plants erect and remove foliage and fruit up to a height of 15-20 cm to avoid moist and stagnant air conditions.
- Collect and destroy the affected fruits regularly.
- Apply pine needle/grass mulch on the field floor to create a barrier between the host and soil borne inoculum.
- With the **onset of monsoon rains, spray the crop with metalaxyl + mancozeb (0.25%)** followed by sprays of either mancozeb (0.25%) or copper oxychloride (0.3%) or Bordeaux mixture (4:4:50) and repeat at 7-10 days interval.

Early blight/ Alternaria leaf spot *Alternaria solani, A. alternata, A. alternata* f.sp *lycopersici*

Tomato



Early blight/ Alternaria leaf spot *Alternaria solani, A. alternata, A. alternata* f.sp *lycopersici*

Early blight/ Alternaria leaf spot *Alternaria solani, A. alternata, A. alternata* f.sp *lycopersici*

Symptoms

- Small, isolated, scattered pale brown spots on the leaf
- Fully developed spots are irregular, brown to dark brown in colour, and with **concentric rings** inside the spot
- Spots coalesce to form large patches resulting in the leaf blight
- Lowest leaves are attacked first and the disease progresses upwards
- In severe attacks the entire plant may be defoliated
- Zonate lesions may also develop on **stems** and petioles, which break at the point of infection
- Brown spots are also seen on **calyx**
- Slightly dark, sunken, round to irregular lesions on fruit at the calyx end

Early blight/ Alternaria leaf spot *Alternaria solani, A. alternata, A. alternata* f.sp *lycopersici*

Disease cycle

PI: Mycelium or conidia **survive in diseased plants debris** and can **persist for one to two years.**

- **Seed borne nature of *A. solani*** has also been reported.
- Primary infection of lower leaves first takes place through conidia formed on crop debris in soil.

SI:

- Conidia developed on primary spots, blown by wind, water and insects to the neighbouring leaves\ plants.

Early blight/ Alternaria leaf spot *Alternaria solani, A. alternata, A. alternata* f.sp *lycopersici*

Epidemiology

- **Dry and warm weather alternating with intermittent rains**
- **Reduction in plant vigour and senescence**
- The optimum temperature for infection of *A. solani* is 28 to 30°C while for *A. alternata* f. sp. *lycopersici* and *A. alternata*, it ranges between 25-30 and 20-25°C, respectively.
- Maximum dispersal of conidia occurs in advanced stages of disease development and in between **9 am and 12 noon**.

Early blight/ Alternaria leaf spot *Alternaria solani, A. alternata, A. alternata* f.sp *lycopersici*

Management

- Maintain plant vigour
- Select healthy seed and treat it with captan (0.3%).
- Collect and destroy the infected plant debris.
- Follow at least two years crop rotation.
- **Remove the foliage particularly in indeterminate type of cvs/hybrids up to 15-20 cm** to avoid moist and stagnant air conditions.
- Spray the crop with chlorothalonil (0.2%) or mancozeb (0.25%) and repeat at 10 to 14 days interval.
- **Early blight resistant F1 hybrids** - Arka Abhed, Arka Samrat, Arka Rakshak, Arka Aditya

Cercospora leaf spot- *Pseudocercospora fuligena*

Small yellow indistinct spots without margins are the early indication for Cercospora leaf spot.

Small brown lesions start to form in the yellow discolored area also.

The yellow spots and the small lesions, later turns to larger necrotic spots with a marked zone on the upper and under side of the leaves.

Cercospora leaf spot- *Pseudocercospora fuligena*

Large necrotic sections on the leaves can be seen in the field.
The disease can **cause severe defoliation of the leaves** under ideal conditions

Cercospora leaf spot can cause large leaf lesions and blighting in tomatoes.

Disease can be a problem under **warm and wet weather conditions.**

Cercospora leaf spot- *Pseudocercospora fuligena*

Symptoms

- Symptoms initially appeared as **yellow spots** on the upper surface (adaxial) of the leaflets, **corresponding to a dark gray fuliginous growth** on the abaxial surface, evolving to irregular necrotic spots and increase in size

- As the disease progress, leaf spots coalesce and become necrotic

- The disease can cause **severe defoliation of the leaves** under ideal conditions

Cercospora leaf spot- *Pseudocercospora fuligena*

Disease cycle

P.I: Mycelium or conidia in infected plant debris

S.I: Air borne conidia

Favourable conditions

Warm, moist weather

Cercospora leaf spot- *Pseudocercospora fuligena*

Management

- Rouging and burning of infected plant debris
- Foliar application of mancozeb@0.25%

Septoria leaf spot (*Septoria lycopersici*)

Symptoms on lower, oldest leaves of transplants in greenhouse.

Small dark, pimple-like fruiting bodies (pycnidia)

Septoria leaf spot (*Septoria lycopersici*)

Lower leaf surface

Septoria leaf spot (*Septoria lycopersici*)

Symptoms observed on plants in the field.

Septoria leaf spot (*Septoria lycopersici*)

Symptoms:

- **Water soaked spots appear** on the lower surface of leaf.
- Later these become more or less circular in outline and show **definite brown coloured margins with grey centre**
- A few **black glistening pinhead sized pycnidia** may be seen in the center of spots.
- **On stem, disease appears as small, slightly elongated, dark spots** containing numerous black pycnidia.
- **Black circular spots** may also be seen on **young as well as mature fruits.**

Septoria leaf spot (*Septoria lycopersici*)

Disease cycle and Epidemiology:

PI : Infected **plant debris** in the form of mycelium or conidia or solanaceous weed hosts debris, horse nettle (*Solanum carolinense* L.)

SI : The conidia are splashed by rain to the lower leaves of the plants.

The spores produced in pycnidia continue the secondary cycle of this disease.

Temperature of **20-25°C with 75-92 %** relative humidity is congenial for disease development

Septoria leaf spot (*Septoria lycopersici*)

Management:

- Destruct infected plant debris and use of clean seed.
- One to two years crop rotation, **trimming of lower leaves and staking of plants in case of indeterminate varieties/ hybrids** have been found useful.
- With the initiation of the disease, spray the crop with carbendazim (0.1%) or mancozeb (0.25%) and repeat at 10 to 14 days interval.

Stemphylium (gray) leaf spot (*Stemphylium lycopersici*)

- **Brown specks scattered** through out the leaf surface is a characteristic symptom
- These specks enlarge become **necrotic and breaks off** from the center.
- Necrotic spots can also merge causing collapse of large areas of the leaf tissue.

Severe necrosis throughout the underside and the margin of the leaves.
This can cause severe defoliation.
When the infection becomes severe leaves of infected plants drops from the plant.

Stemphylium (gray) leaf spot (*Stemphylium lycopersici*)

Stemphylium (gray) leaf spot (*Stemphylium lycopersici*)

Symptoms

- Small **brownish black specks** on the underside of leaves
- Later these specks develop on both leaf surfaces into grayish **brown, glazed lesions of 3 mm** or less in size
- On older leaves lesions dry up, **crack and the centers drop out leaving a shot hole appearance.**
- **Yellowing of the leaf** occurs followed by defoliation of the plant

Stemphylium (gray) leaf spot (*Stemphylium lycopersici*)

Disease cycle

P.I: Mycelium or conidia in infected plant debris

S.I: Air borne conidia

Favourable conditions

Warm, moist weather

Stemphylium (gray) leaf spot (*Stemphylium lycopersici*)

Management

- Rouging and burning of infected plant debris
- Foliar application of mancozeb@0.25%

Leaf spots /blights - *Alternaria, Septoria* and *Stemphylium*

Survival and spread : Seed borne and air borne conidia

Management

- Select the seeds from healthy fruits and treat the seed with captan or thiram@3g/kg seed
- Avoid close planting.
- Destroying the infected leaves
- Avoid wetting leaves while watering
- Watering should be early in the day to allow foliage to dry quickly.
- Spray Mancozeb@0.25% or Copper oxy chloride @0.3%

Anthracnose- (*Colletotrichum gloeosporioides)*

Anthracnose- (*Colletotrichum gloeosporioides)*

Symptoms

- **More pronounced on the ripen fruits**
- **Leaves**-Characterized by small **circular brown lesions surrounded by yellow halo**
- **Fruits**-Gets infected when **green in color but symptoms will appear when begins to ripe**.
- Lesions enlarge & becomes **sunken with concentric ring markings.**
- Centre of the spot is usually **tan coloured and with black dots** by the time it comes maturity.
- **Stems**-Lesions on stems are **brown & elongated**.

Anthracnose - (*Colletotrichum gloeosporioides)*

Management

- Mulching in between the rows & spraying of chemicals like carbendazim 0.1%.

Fusarial wilt (*Fusarium oxysporum* f.sp. *lycopersici*)

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Fusarial wilt (*Fusarium oxysporum* f.sp. *lycopersici)*

Symptoms

- The disease first appears in the field as yellowing of the lower leaves most commonly at the time of flowering stage.
- The affected leaves die prematurely and the symptoms continue to appear on successively younger leaves.
- **One or more branches may be affected** while the others remain symptomless.
- Browning of the walls of the vascular bundles follows rather promptly, and is best seen by cutting of stem or petiole and examining a cross section.

Fusarial wilt (*Fusarium oxysporum* f.sp. *lycopersici)*

Disease cycle

PI: As mycelium and spore forms especially **chlamydospores** in the infected plant debris and in the soil
SI: It spreads over small distances by means of **water and contaminated farm equipments**, and over long distances, primarily in **infected transplants or in the soil** carried with them.

Usually, once an area becomes infested with *Fusarium*, it **remains so indefinitely.**

Fusarial wilt (*Fusarium oxysporum* f.sp. *lycopersici)*

Epidemiology:

- The optimum soil temperature for disease development varies from 25-31°C.
- Hot, dry weather favour wilt development.
- The disease is **more severe in acidic soils (5.6 to 6.5 pH) than in alkaline ones.**
- **Presence of root knot nematodes** and monoculture enhance the disease further.
- **Low nitrogen and high potassium levels in soil predisposes the plant to this disease**.

Fusarial wilt (*Fusarium oxysporum* f.sp. *lycopersici)*

Management

- Follow careful crop rotation and use of healthy seed.
- Treat the seed with carbendazim (0.2%).
- Avoid movement of water from infected to healthy plants.
- Combined application of **inorganic fertilizers and organic manures** is considered effective in reducing disease incidence.
- **Application of phosphate fertilizers and nitrate fertilizers** help in reducing the wilt incidence whereas **ammonium fertilizers** increase the disease development.
- Drench the affected plants with carbendazim (0.1%).
- Soil application of biocontrol agents- *Trichoderma viride* *Pseudomonas fluorescens* can also reduce infection

Collar rot- *Sclerotium rolfsii* (Tel *Athelia rolfsii*)

Yellowing of lower leaves and death of the whole plant

Collar rot- *Sclerotium rolfsii* (Tel *Athelia rolfsii*)

White fungal growth on the root

Collar rot- *Sclerotium rolfsii* (Tel *Athelia rolfsii*)

White fungal growth near the collar region and sclerotial bodies

Collar rot- *Sclerotium rolfsii* (Tel *Athelia rolfsii*)

Collar rot- *Sclerotium rolfsii* (Tel *Athelia rolfsii*)

Symptoms

- Very destructive soil borne disease
- Severe in warm weather.
- Sudden wilting of the plants
- Base of the **stem near soil line** & **fruits** that come in contact with soil shows tissue necrosis.
- Mustard like sclerotial bodies of the fungus is observed

Collar rot- *Sclerotium rolfsii* (Tel *Athelia rolfsii*)

Disease cycle:

- Fungus soil borne, survive as saprophyte on crop debris and as sclerotia
- **High temperature and humidity** favour the disease development.
- In the plains of India, because the **lower leaves become covered with soil resulting in a bridge** of dead tissue which furnishes an ideal medium for initiating pathogenesis.

Collar rot- *Sclerotium rolfsii* (Tel: *Athelia rolfsii*)

Management

- Sanitation/ removal of diseased crop debris
- Crop rotation with non-host crops like cereals.
- **Deep summer ploughing**
- Soil solarisation and amendment with neem cake
- **Seed treatment with carbendazim at 2g/kg**
- Use disease free planting material.
- Recommended dose of fertilizer application
- Application of organic manures fortified with biocontrol agents (*T. viride and P. fluorescens* @ 1-2 kg/ac)

Collar rot- *Sclerotium rolfsii* (Tel: *Athelia rolfsii*)

Management contd…

- Drenching the soil with 0.1% Carbendazim or 0.3% COC or Drenching wit chestnut compound (copper sulphate 2 parts + Ammonium carbonate 11parts) at 3 gm/l